THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home: Earth, a Unique Life-sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A ray strikes a plane mirror making an angle of 25 degrees with the mirror. The angle of reflection is: 1

(a) 50 degrees **(b)** 65 degrees **(c)** 25 degrees **(d)** 155 degrees

2. A vertical stick of 1 m casts a shadow of 1 m long at some time. The Sun's elevation at that time is about: 1

(a) 60 degrees **(b)** 30 degrees **(c)** 45 degrees **(d)** 90 degrees

3. In the food chain: phytoplankton, zooplankton, small fish, big fish, the secondary consumer is: 1

(a) small fish **(b)** phytoplankton **(c)** big fish **(d)** zooplankton

4. If the radius of the Earth is about 6400 km, the thickness of the crust (about 35 km) is: 1

(a) about 25 per cent of the radius **(b)** equal to the radius
(c) about 50 per cent of the radius **(d)** less than 1 per cent of the radius

5. A concave mirror of focal length 10 cm has a radius of curvature: 1

(a) 40 cm **(b)** 20 cm **(c)** 5 cm **(d)** 10 cm

6. A lunar year consists of 12 lunar months of 29.5 days each. The number of days is: 1

(a) 354 **(b)** 360 **(c)** 365 **(d)** 348

7. Which gas do aquatic plants release in the daytime that fish need? 1

(a) methane **(b)** nitrogen **(c)** oxygen **(d)** carbon dioxide

8. If air contains 21 per cent oxygen, in 500 L of air the volume of oxygen is about: 1

(a) 78 L **(b)** 105 L **(c)** 21 L **(d)** 210 L

9. A convex lens converges a parallel beam to a point called the: 1

(a) principal focus **(b)** pole **(c)** optical centre **(d)** axis

10. The Moon's waxing crescent is seen in the: 1

(a) west after sunset **(b)** south at noon
(c) east at midnight **(d)** north at dawn

11. In a pond, the amount of dissolved oxygen falls when: 1

(a) plants grow **(b)** it is cleaned
(c) organic waste rots in it **(d)** it rains

12. The percentage of Earth's water that is fresh water is about: 1

(a) 50 per cent **(b)** 30 per cent **(c)** 3 per cent **(d)** 97 per cent

13. The angle between the incident and the reflected ray is 80 degrees. The angle of incidence is: 1

(a) 10 degrees **(b)** 160 degrees **(c)** 80 degrees **(d)** 40 degrees

14. At noon the shadow of a stick is directly towards the north (in India). The Sun then is: 1

(a) to the north **(b)** below the horizon

(c) overhead **(d)** to the south of the observer

15. A sudden increase in the population of rabbits in a grassland may occur if: 1

(a) grass is removed **(b)** owls are added

(c) rains are heavy **(d)** foxes are removed

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. A ray of light makes an angle of 30 degrees with a plane mirror. What is the angle between the incident and reflected rays? 2

17. A stick of height 2 m casts a shadow of 2 m at 10 am. What is the angle of the Sun above the horizon? 2

18. Why are there fewer top carnivores than herbivores in an ecosystem? 2

19. Why is the inner core solid though the temperature is very high? 2

20. Why does the image formed by a convex mirror always appear smaller? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A ray falls on a plane mirror at 40 degrees to the normal. The mirror is turned by 10 degrees keeping the incident ray fixed. Find the new angle of incidence and the angle by which the reflected ray turns. 3

22. A student marks the tip of the shadow of a 1 m stick at 9 am, 12 noon and 3 pm. Describe the pattern and the shadow lengths relationship. 3

23. Construct a food chain of four organisms in a pond and identify the trophic level of each. 3

24. The Earth's radius is 6400 km. The crust is 35 km, mantle 2900 km. Find the total thickness of the core. 3

OR

Why does the driver of a vehicle prefer convex rear-view mirrors? Mention any drawback. 3

25. Compare the Gregorian, Islamic and Hindu calendars on basis of the Sun, Moon and year length. 3

OR

State how excessive hunting of wolves in a forest would affect deer and the forest vegetation. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe an experiment to verify the laws of reflection. State the apparatus, procedure, observations and conclusion. 5

OR

Describe an activity to find the time of local noon using the shadow of a stick and explain how you find north-south. 5

27. A grassland has grass, grasshoppers, mice, frogs, snakes and hawks. Draw the food web in words, list the chains, and predict the effect of removing the snakes. 5

OR

Describe the structure of the Earth's interior with the thickness (approximately), state and composition of each layer, and how it is studied. 5

END OF QUESTION PAPER **Total: 50 Marks**

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